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Q1

The artery carries blood away from the heart. The aorta is the largest artery and it carries oxygenated blood to the body.

Q2

The chloroplast is the organelle involved in photosynthesis. It contains chlorophyll which traps sunlight.

Q5

Transpiration is the loss of water vapour from the leaves of a plant through the stomata. It pulls water up from the roots. Two factors that increase the rate are high temperature and windy conditions.

Q3

Chloroplasts contain chlorophyll a and chlorophyll b which absorb light. The two stages are the light reaction which happens in the thylakoid and makes ATP, and the Calvin cycle in the stroma which fixes carbon dioxide into glucose.

Q7 a

Plant B was kept in dim light so it could not make enough food. It grew tall and pale because it was stretching towards the light and had less chlorophyll.

Q7 b

Move Plant B to a bright window so it gets enough sunlight.

Q4

Blood enters the right atrium from the vena cava. It goes through the tricuspid valve into the right ventricle, then out the pulmonary valve to the lungs. It comes back to the left atrium, through the bicuspid valve into the left ventricle and out through the aortic valve into the aorta.

Note

Remember to revise the diagram of the nephron before the next test.

Q8

Tidal volume = 0.5 L, dead space = 0.15 L. So fresh air per breath = $0.5 - 0.15 = 0.35$ L.

Continuing Q8: alveolar ventilation = $0.35 \times 12 = 4.2$ L per minute.